

2.1 Ports & Protocols

20 / 21	FTP – file transfer
22	SSH / SFTP – secure shell
23	Telnet – unencrypted
25	SMTP – send mail
53	DNS – name to IP
67 / 68	DHCP – 67 server, 68 client
80 / 443	HTTP / HTTPS (443 = TLS)
110 / 143	POP3 / IMAP
161 / 162	SNMP – monitor
80->443 & 23->22 are the plain->secure pairs. POP pulls, IMAP syncs.	

3.4 RAID Levels

0 – stripe	min 2 · survives 0 (speed only)
1 – mirror	min 2 · survives 1 drive
5 – stripe+parity	min 3 · survives 1 drive
6 – dbl parity	min 4 · survives 2 drives
10 – 1+0	min 4 · survives 1 per mirror
0 = zero protection · 5 survives 1 · 6 survives 2 · 10 = best of both, costs most.	

2.3 Wi-Fi & Bands

a / b / g	5 / 2.4 / 2.4 GHz – 54 / 11 / 54M
n · Wi-Fi 4	2.4 / 5 GHz – 600M
ac · Wi-Fi 5	5 GHz – ~7G
ax · Wi-Fi 6/6E	2.4 / 5 / 6 GHz – 9.6G
2.4 = range · 5 = speed · 6 = clean (6E). Channels 1/6/11. WPA3 > WPA2 > never WEP.	

3.1 Cables & Connectors

RJ45 / RJ11	ethernet / phone
Cat 5e	1G · 100m
Cat 6 / 6a	10G · 55m / 100m
Coax RG-6	F-type · cable/sat
Fiber	LC · SC · ST connectors
USB-C / TB	up to 40 Gbps

2.5 IP Addressing

Class A	10.0.0.0 – /8
Class B	172.16-31 – /16
Class C	192.168 – /24 (254 hosts)
APIPA	169.254.x – no DHCP
Loopback	127.0.0.1
APIPA 169.254.x = DHCP failed to answer. Loopback 127.0.0.1.	

- Acronyms – Net & Hardware

NAT / PAT	private to public IP
VLAN	logical network segment
DHCP / DNS	auto-IP / name resolution
UEFI / BIOS	firmware (UEFI = GPT + secure boot)
TPM	hardware security chip (BitLocker)
PoE	power over ethernet
SOHO	small office / home office
PXE	boot from the network

REAL TALK

The cert proves you can **pass**. The job is something else. My first night as a SysAdmin a workstation wouldn't join the domain – and the “fix” wasn't a trick from a practice test. I noticed its IP wasn't even on the network; wrong subnet, couldn't reach the DC. **Labbing is what taught me to look there first.** Memorize this sheet to pass – then go break things in a homelab until the “why” is yours. That's the part that gets hired. – **James**

5.2 POST Beep Codes (BIOS-Vendor Specific)

RULE #1	Beep meanings differ by BIOS vendor (AMI vs Award/Phoenix). ALWAYS check the mobo/BIOS doc.
1 beep / no beep	Normal POST pass (most boards). No beep can also mean dead speaker/PSU.
AMI: 1 long + 2 short	Video / GPU failure (no display adapter).
AMI: 1 long + 3 short	Memory failure above 64KB (reseat/replace RAM).
AMI: continuous	RAM or power issue (reseat/replace RAM).
Award/Phoenix: 1 long + 2 short	Video card error.
Phoenix: beep patterns	Coded in groups (e.g. 1-3-1) = memory; consult Phoenix chart.
Repeating short beeps	Commonly RAM not seated/bad, or power problem.
No beeps + no display	PSU, CPU, or motherboard dead; check power before parts.

Beep codes are a starting clue, not gospel - map them to YOUR board's manual.

5.3 No-POST / No-Power First Checks + HW Failure Signs

Power FIRST	Wall outlet, PSU switch (I/O), voltage selector (115/230), power cable seated.
Front-panel header	Confirm PWR_SW jumper wired right; test by shorting power pins.
Reseat the basics	RAM, GPU, CPU power (4/8-pin EPS), 24-pin ATX, drive cables.
Strip to minimum	1 RAM stick, onboard/1 GPU, no peripherals; add back one at a time.
Clear CMOS	Pull battery / move jumper to reset bad BIOS settings or OC.
Mobo POST LEDs	Debug LEDs (CPU/DRAM/VGA/BOOT) pinpoint the failing stage.
Continuous beeps / no display	Bad RAM or GPU - reseat/swap known-good.
Burning smell / smoke	KILL POWER NOW. Likely PSU or VRM; do not re-power.
Clicking / grinding drive	Failing HDD (click of death). Back up immediately; replace.

Order: power -> reseat -> minimal config -> swap known-good. Smell/smoke overrides all - cut power first.

3.8 Printer Types: Trait + Consumable

Laser	EP process, fast high-volume crisp text; consumable = toner cartridge (+ drum, fuser)
Inkjet	Sprays liquid ink droplets, great photo/color; consumable = ink cartridges + paper
Thermal	Heated element on special heat-sensitive paper (receipts); consumable = thermal paper
Impact / Dot-Matrix	Print head pins strike inked ribbon; only one that does carbon multipart forms; consumable = ribbon
3D	Builds object in layers; FDM melts filament, resin uses liquid; consumable = filament or resin

Impact = the multipart/carbon-copy answer. Thermal = receipts.

3.8 Laser Imaging Process – EXACT 7-Step Order

1. Processing	Raster image built in printer memory (RIP); page rendered to bitmap
2. Charging	Primary/conditioning corona or charge roller applies uniform negative charge to drum
3. Exposing	Laser writes the image - discharges dots to a lower charge (latent image)
4. Developing	Charged toner attracted to the exposed (written) areas of the drum
5. Transferring	Transfer corona/roller moves toner from drum to paper
6. Fusing	Heat + pressure rollers melt toner permanently into paper
7. Cleaning	Blade scrapes residual toner; erase lamp neutralizes drum for next page

Mnemonic: 'Please Charge Every Day To Fix Computers.' Graded exam fact - memorize this order cold.

5.6 Common Printer Problems + Fixes

Ghosting (faint repeat)	Bad drum, worn fuser, or failing erase lamp; replace drum/imaging unit or fuser
Vertical white streaks	Clogged/empty toner or debris on transfer roller; shake or replace toner, clean path
Vertical lines/smudges (laser)	Scratched drum or dirty corona wire; replace drum, clean corona
Paper jams	Worn pickup/separation rollers, wrong/damp paper, debris; clear path, fresh dry paper, replace rollers
Faded / light print	Low toner or low density setting; replace toner, raise print density, check economode
Smudging / not fused	Failed fuser (heat) or wrong media type setting; replace fuser, set correct paper type
Garbled output	Wrong/corrupt driver or bad cable/connection; reinstall correct driver, reseat cable, clear queue

Streaks = consumable (toner/drum); jams = rollers/paper; garbled = driver/cable.

1.1 Windows Editions – Key Differences

Home	Consumer tier. NO domain join, NO Group Policy editor (gpedit), NO BitLocker, NO Remote Desktop HOST (client only).
Pro	Adds domain join, gpedit/Group Policy, BitLocker, RDP host, Hyper-V. Business baseline.
Pro for Workstations	Pro + ReFS filesystem, persistent memory, faster file sharing (SMB Direct), up to 4 CPUs & 6TB RAM.
Enterprise	Volume-license only. Adds AppLocker, BranchCache, DirectAccess, Credential Guard, Windows To Go (legacy).
RAM cap (64-bit)	Home 128GB; Pro 2TB; Pro Workstations 6TB; Enterprise 6TB.
Domain vs Workgroup	Domain join = Pro and up only; Home is workgroup/Azure AD only.
Exam favorite: BitLocker, RDP host, gpedit, and domain join are all the Pro upgrade line.	

1.4 / 1.8 File Systems – Traits & Size Limits

NTFS	Windows default. Journaling, permissions/ACLs, encryption (EFS), compression, quotas. Max file/vol effectively huge (256TB+).
FAT32	Legacy/cross-platform. Max FILE size 4GB; max volume 32GB (Windows format). No permissions/journaling.
exFAT	Flash drives / large SD cards. No 4GB file limit; cross-platform (Win/macOS). No journaling or permissions.
ext4	Linux default. Journaling, large files/volumes (16TB file / 1EB vol). Not natively read by Windows.
APFS	macOS/iOS default (replaced HFS+). Snapshots, cloning, encryption, SSD-optimized.
Pick rule	Win internal = NTFS; cross-OS flash = exFAT; old/small/compat = FAT32.

Memory hook: FAT32 = 4GB file / 32GB volume ceiling – the classic ‘won’t copy a movie’ answer.

1.2 / 1.3 Windows CLI, Utilities & MMC Snap-ins

sfc /scannow	System File Checker – scans & repairs protected Windows system files.
chkdsk /f /r	Checks disk for errors; /f fixes, /r locates bad sectors & recovers data.
dism	Deployment Image Servicing – repair component store: /Online /Cleanup-Image /RestoreHealth.
diskpart	CLI disk partitioning – create/delete/format partitions, convert MBR↔GPT.
gpupdate /force / gpresult	gpupdate refreshes Group Policy; gpresult shows RSoP (which policies applied).
taskmgr / msconfig	Task Manager (processes/startup/perf); msconfig = System Config (boot, services, tools).
regedit / eventvwr	Registry Editor; Event Viewer (system/app/security logs).
.msc snap-ins	devmgmt (Device Mgr), services.msc (Services), perfmon (Performance Monitor).

DISM vs SFC order: run SFC first; if it can’t fix, DISM /RestoreHealth repairs the store, then re-run SFC.

1.9 / 1.1 OS Install & Boot Concepts

Clean install	Wipes drive, fresh OS. No app/setting carryover. Best for corruption/repurpose.
In-place upgrade	Installs over existing OS; KEEPS apps, files, settings. Same/compatible edition path.
Image deploy	Push a prebuilt image (e.g. via DISM/imaging) to many machines fast. Enterprise rollouts.
GPT vs MBR	GPT: >2TB disks, up to 128 partitions, requires UEFI. MBR: ≤2TB, 4 primary partitions, legacy BIOS.
UEFI vs Legacy	UEFI: GPT, Secure Boot, fast boot, GUI/mouse. Legacy/BIOS: MBR only, no Secure Boot.
Boot order	Set in UEFI/BIOS – which device is tried first (USB/optical/SSD); key for install media or PXE.
Secure Boot	UEFI feature blocking unsigned bootloaders; Win11 requirement (with TPM 2.0).
Win11 install gotchas: requires UEFI + Secure Boot + TPM 2.0 + GPT – a frequent troubleshooting item.	

3.1 Windows Symptoms & Fixes

BSOD / stop error	Note exact stop code; reboot to Safe Mode; roll back driver/update; run SFC + chkdsk; check RAM/overheating
Slow profile load	Bloated/corrupt roaming profile, slow GPO/login scripts, mapped-drive timeouts; clear temp, repair profile, check network
No boot	Boot to WinRE/install media; bootrec /fixmbr, /fixboot, /scanos, /rebuildbcd; chkdsk; check boot order/media
Service won't start	services.msc / Event Viewer for error; verify dependencies + startup type; correct logon account/credentials
System time drift	Dead CMOS/RTC battery (also resets BIOS), wrong time zone, w32time/NTP not syncing; replace battery, re-sync
Low disk space	Disk Cleanup / Storage Sense, clear temp + old restore pts, uninstall apps, move/expand volume

3.1 covers Windows OS symptoms – always capture the exact error before changing anything.

3.1 Common BSOD Stop Codes

IRQL_NOT_LESS_OR_EQUAL	Usually a faulty/incompatible driver; can also be bad RAM
PAGE_FAULT_IN_NONPAGED_AREA	Bad RAM or a driver/service referencing invalid memory
SYSTEM_SERVICE_EXCEPTION	Driver or system file fault; often after a driver/update change
UNMOUNTABLE_BOOT_VOLUME	Corrupt filesystem / failing disk / bad boot config; run chkdsk + bootrec
CRITICAL_PROCESS_DIED	Corrupt system file or driver killed a critical process; run SFC/DISM
Key caveat	A stop code points to a category, NOT one cause – same code can mean driver, RAM, disk, or firmware

Don’t treat a code as a single diagnosis; corroborate with Event Viewer, recent changes, and hardware tests.

2.4 Malware Types

Virus	Attaches to a host file; runs/spreads when user executes it
Worm	Self-replicates across networks; needs no host or user action
Trojan	Disguised as legit software; opens a backdoor once run
Ransomware	Encrypts files/locks system, demands payment to restore
Rootkit	Hides at OS/kernel/boot level for stealthy admin access; hard to detect
Keylogger	Records keystrokes to steal passwords/credentials
Spyware	Covertly monitors activity, harvests data, tracks habits
Cryptominer	Hijacks CPU/GPU to mine cryptocurrency; spikes resource use
Boot sector	Infects MBR/boot record; loads before OS, survives a simple reinstall
Rootkit + boot sector load before the OS, so they evade normal AV.	

2.5 Social Engineering + Threats

Phishing	Mass fraudulent email/link to steal creds or drop malware
Spear phishing	Phishing targeted at a specific person/org with personalized bait
Whaling	Spear phishing aimed at high-value execs (CEO/CFO)
Vishing	Voice/phone-call phishing
Smishing	SMS/text-message phishing
Tailgating	Following an authorized person through a secure door
Shoulder surfing	Watching screen/keypad to steal info
Dumpster diving	Digging through trash for sensitive docs/data
Impersonation	Posing as IT/staff/authority to gain trust and access
Vishing=voice, Smishing=SMS, Whaling=execs. Don't mix them up.	

2.1 / 2.3 Auth Best Practices + Wireless/Encryption

MFA	2+ factors: know (PW) / have (token) / are (biometric)
Biometrics	Fingerprint, face, retina; 'something you are' factor
Least privilege	Grant only the minimum access needed for the role
Strong password policy	Length, complexity, expiration, history, lockout
WPA3 + AES	PREFERRED; SAE handshake, strongest current Wi-Fi security
WPA2 + AES	Acceptable fallback; CCMP/AES cipher
Avoid WEP	Broken/legacy; cracked in minutes
Avoid TKIP	Deprecated cipher (old WPA); use AES instead
Best to worst: WPA3>WPA2(AES)>WPA(TKIP)>WEP.	

MALWARE REMOVAL · SOHO 7-Step Malware Removal Procedure (EXACT ORDER)

2.6

- 1 Investigate/verify** – Investigate and verify malware symptoms
- 2 Quarantine** – Quarantine the infected system (disconnect from network)
- 3 Disable Restore** – Disable System Restore in Windows
- 4 Remediate** – Update anti-malware; scan + remove (Safe Mode / preinstall env)
- 5 Schedule + update** – Schedule scans and run updates
- 6 Enable Restore** – Re-enable System Restore and create a restore point
- 7 Educate user** – Educate the end user

TROUBLESHOOTING METHOD 6-Step Troubleshooting Methodology

Best Practice

- 1 Identify problem** – Gather info, question user, ID changes; back up before making changes
- 2 Theory of cause** – Establish probable cause (question the obvious); research if needed
- 3 Test the theory** – Test to confirm; if confirmed plan next steps, if not form a new theory or escalate
- 4 Plan of action** – Establish plan to resolve and implement the solution
- 5 Verify** – Verify full system functionality and, if applicable, apply preventive measures
- 6 Document** – Document findings, actions, and outcomes